

24PY901
Quantum Principles, Structures and Computing
Module 1 – Unit 2

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Assignment 3: Linear system formulation ¹

Problems from CSE, ECE, VLSI, and BT

References:

1. Griffiths. Appendix on Linear Algebra
2. World Wide Web

Instructions

- This assignment focuses only on **mathematical modeling** in general and **linear modeling** in particular.
- You are NOT required to solve the systems.
- Your task is to:
 - Define variables clearly.
 - Convert the scenario into linear equations.
 - Represent the system in matrix form $AX = B$.
 - Identify dimensions of matrices involved.
- The expectation level corresponds to **TRL 2 (Technology Formulation)**.

¹Level of difficulty: TRL 2. TRL, which stands for Technology Readiness Level, is a scale developed by NASA to measure maturity of projects, especially for space missions. The scale runs from TRL 1 (basic research) to TRL 9 (fully proven in operational missions). For the present assignment, TRL 2 stands for “Technology concept formulated”.

Problem 1: Resource Allocation in Cloud CPU Task Scheduling (CSE / Interdisciplinary)

Background Context

Modern computing systems such as cloud servers, data centers, and high-performance computing clusters must run multiple types of tasks simultaneously.

Each task consumes system resources such as:

- RAM (memory)
- CPU cores (processing capability)
- Storage I/O bandwidth

Since these resources are limited, system administrators must determine how many tasks of each type can be executed without exceeding hardware constraints.

This type of modeling is fundamental in:

- Cloud computing platforms (AWS, Azure, Google Cloud)
- Virtual machine scheduling
- Container orchestration (e.g., Kubernetes)
- AI model training clusters
- Embedded real-time systems

Engineering Scenario

A cloud server executes three types of processes:

- Data Processing (e.g., database queries)
- Rendering (e.g., graphics/video processing)
- Encryption (e.g., secure communication tasks)

Each task consumes the following resources:

Task Type	RAM (GB)	CPU Cores	I/O Units
Data Processing	2	1	3
Rendering	4	2	2
Encryption	1	2	1

The total available resources in the server are:

- 40 GB RAM
- 20 CPU cores
- 30 I/O units

The goal is to model how many tasks of each type can be run without exceeding available system resources.

Interpretation

- Each resource constraint gives rise to a linear equation.
- The problem represents a system of linear equations.
- The coefficient matrix represents how tasks consume resources.
- The solution vector represents number of tasks of each type.

This is an example of modeling a real-world allocation problem using linear algebra before applying optimization techniques.

Task (TRL 2 – Technology Formulation Only)

You are required to:

1. Define decision variables for the number of each task type.
2. Formulate linear equations representing:
 - RAM constraint
 - CPU constraint
 - I/O constraint
3. Express the system in matrix form $AX = B$.
4. Identify:
 - Dimensions of A , X , and B
 - Whether the system is square
 - Whether multiple feasible allocations may exist
5. State the rank condition required for a unique solution.

Extension (Conceptual Only)

Suppose an additional constraint is added:

- The total number of tasks must be exactly 15.
- How does this modify the matrix system?
- Does the system become square?
- Under what condition would the solution be unique?

(No calculations required.)

Problem 2: Signal Mixing in a Communication Transmitter (ECE)

Background Context

In modern communication systems (Wi-Fi, 5G, satellite links, Bluetooth, etc.), signals are often combined before transmission. This process is called **linear signal mixing**.

For example:

- In Multiple Input Multiple Output (MIMO) systems
- In antenna array beamforming
- In audio signal mixing
- In sensor data fusion systems

A transmitter may generate multiple baseband signals. These signals are combined using linear amplifiers or mixers before being sent to antennas.

Mathematically, the transmitted signals are linear combinations of input signals. This means each output signal is formed by scaling and adding the inputs.

Engineering Scenario

Suppose three baseband signals:

$$S_1, \quad S_2, \quad S_3$$

represent:

- S_1 : Voice data stream
- S_2 : Video data stream
- S_3 : Control signaling data

These are combined to generate two transmitted signals T_1 and T_2 according to hardware mixing coefficients determined by circuit design:

$$T_1 = 2S_1 + 1S_2 + 3S_3$$

$$T_2 = 1S_1 + 4S_2 + 2S_3$$

During testing, engineers measure:

$$T_1 = 20 \text{ volts}, \quad T_2 = 25 \text{ volts}$$

The objective is to model how input signals relate to transmitted outputs.

Interpretation

- Each coefficient represents gain applied by the mixing circuit.
- This is a linear transformation from a 3-dimensional signal space to a 2-dimensional transmitted space.
- The system can be interpreted as a matrix transformation:

$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

This modeling approach is fundamental in:

- MIMO systems
- Beamforming
- Signal demixing
- Blind source separation
- Linear algebra-based communication theory

Task (TRL 2 – Technology Formulation Only)

You are required to:

1. Define the unknown signal variables clearly.
2. Write the system in matrix form $AX = B$.
3. Identify:
 - Dimensions of A , X , and B
 - Whether the system is square, underdetermined, or overdetermined
 - The rank condition required for unique recovery of input signals
4. Interpret what it means physically if:
 - $\text{Rank}(A) < 2$
 - The system has infinitely many solutions
 - No solution exists

Extension (Conceptual Only)

Suppose instead that there are 3 transmitted outputs and 3 input signals. How would the matrix structure change?

Under what condition can the input signals be uniquely recovered?

Which linear algebra concept determines this?

(No calculations required.)

Problem 3: Transistor Budget Optimization in VLSI Chip Design (ECE-VLSI)

Background Context

Modern digital integrated circuits (ICs) are built using millions or billions of transistors. However, during early-stage chip design, engineers must carefully allocate available transistors among different logic gates while respecting:

- Total transistor availability (area constraint)
- Total power consumption (thermal constraint)
- Functional design requirements

In CMOS technology, different logic gates require different numbers of transistors. For example:

- A 2-input NAND gate requires 4 transistors.
- A 2-input NOR gate requires 4 transistors.
- A 2-input XOR gate requires 8 transistors.

Additionally, each gate consumes dynamic power when switching:

- NAND gate: 2 mW
- NOR gate: 3 mW
- XOR gate: 5 mW

Assume a small digital processing module is being designed under the following constraints:

- Maximum available transistors = 4000
- Maximum allowable total power = 2500 mW

The design team must determine how many NAND, NOR, and XOR gates can be included without exceeding transistor and power limits.

Engineering Interpretation

- Transistor count constraint represents chip area limitation.
- Power constraint represents heat dissipation and battery limitations.
- The allocation problem can be modeled using linear equations.

This type of modeling is used in:

- ASIC design
- FPGA resource estimation
- Low-power embedded systems
- Early-stage chip architecture planning

Task (TRL 2 – Technology Formulation Only)

You are required to:

1. Define decision variables:
 - Number of NAND gates
 - Number of NOR gates
 - Number of XOR gates
2. Formulate the transistor constraint as a linear equation.
3. Formulate the power constraint as a linear equation.
4. Represent the system in matrix form $AX = B$.
5. State the dimensions of A , X , and B .
6. Identify whether the system is:
 - Square
 - Underdetermined
 - Overdetermined
7. Comment (without solving) whether multiple feasible allocations may exist.

Extension (Conceptual Only)

Suppose an additional constraint is introduced:

- The design must include exactly 200 XOR gates.
- How does this change the system of equations?
- Does the system become square?
- What linear algebra concept determines whether a unique solution exists?

(No calculations required.)

Problem 4: Gene Expression Modeling in a Cellular Pathway (Biotechnology)

Background Context

Inside a biological cell, genes produce messenger RNA (mRNA), which in turn leads to protein production. However, gene regulation networks are interconnected — the expression level of one gene can influence multiple proteins, and each protein may depend on several genes.

In many biotechnology and systems biology studies, the relationship between gene expression levels and protein concentrations is approximated as a **linear model** for small perturbations around an operating point. This is commonly used in:

- Drug response modeling
- Synthetic biology circuit design
- Cancer biomarker studies
- CRISPR-based gene activation analysis

Assume three genes G_1, G_2, G_3 are experimentally activated in a controlled environment. Their expression levels (measured in normalized mRNA units) influence the concentration of three proteins P_1, P_2, P_3 (measured in micrograms/mL).

Based on prior laboratory calibration, scientists determined that protein production follows approximately linear relationships:

$$P_1 = 3G_1 + 2G_2 + 1G_3$$

$$P_2 = 1G_1 + 4G_2 + 2G_3$$

$$P_3 = 2G_1 + 1G_2 + 3G_3$$

In a new experiment, measured protein concentrations are:

$$P_1 = 50, \quad P_2 = 60, \quad P_3 = 55$$

Interpretation

- Each coefficient represents regulatory influence strength.
- For example, coefficient “3” in P_1 equation means gene G_1 strongly contributes to protein P_1 production.
- This system models how gene activity maps to observable protein outputs.

Task (TRL 2 – Technology Formulation)

You are required to:

1. Clearly define the unknown variables.
2. Write the system in matrix form $AX = B$.
3. Identify:
 - Dimensions of A , X , and B
 - Whether the coefficient matrix is square
 - Conditions under which a unique solution exists
 - What linear algebra concept (determinant, rank, invertibility, linear independence) determines solution feasibility
4. Interpret what it means biologically if:
 - $\det(A) = 0$
 - $\text{Rank}(A) < 3$
 - The system has infinitely many solutions

Extension (Conceptual Only)

Suppose instead that there are 5 proteins but still only 3 genes. How would the matrix dimensions change? Would the system be overdetermined or underdetermined? What linear algebra tool would be appropriate in such a case? (No calculations required.)